

lowest and expected returns were highest. The CalPERS board did not establish a formal rebalancing policy until 2009.

The opposite problem is that boards are often attracted to asset classes or strategies that have done well and enter after a period of strong gains when future expected returns are low. This also tends to generate pro-cyclical investing. As real estate returns were heating up during the mid-1990s, CalPERS followed aggressively and ramped up its real estate allocations from a 5% low in 2005 to a peak of more than 9% in 2008—just as real estate was crashing. In 2001, CalPERS' loan-to-value ratio in its real estate portfolio was only 19%. By 2004, the leverage CalPERS was using in its real estate transactions had reached 41% and was projected to reach 50%. In addition to juicing its real estate investments with more leverage, the board jumped on the real estate band wagon and delegated decisions to outside partners so that deals would not get caught up in "red tape." But CalPERS retained the risk. CalPERS only began an external due-diligence process for its real estate investments in January 2008, which was after the market had peaked and many real estate investments had already soured.

Joseph Dear, CIO of CalPERS, described CalPERS' real estate investments as "a disaster."³⁷ CalPERS' real estate holdings lost over 70% of their value between 2008 and 2010. Some of the individual deals are sorry stories indeed. In New York, CalPERS lost \$500 million after lenders took control of the Stuyvesant Town-Cooper Village property in which it had foolishly invested. In 2005, CalPERS bought 9,000 lots of residential housing in Mountain House, California, from Trimark Communities. By 2008, CalPERS was forced to value the deal at negative \$305 million (an "asset" with a negative value!) due to the losses suffered plus the interest it paid on the leverage. That same year, Mountain House was declared the most "underwater" community in America.³⁸ All of this pro-cyclicality was due to poor judgment, lack of risk controls, and a failure to set risk boundaries: these shortcomings created the worst possible alignments between CalPERS and its real estate investment advisors. In addition, complexity and high costs obscured CalPERS' true underlying factor risk exposure.

Norway's sovereign wealth fund has so far had the opposite experience. It was the largest buyer of equities in the fourth quarter of 2008 and it was able to cling to its rebalancing rule while others abandoned theirs. Norway had decided on a process for rebalancing, rather than having a board make rebalancing decisions.³⁹ But Norway also experienced poor active returns during 2008 and 2009. At the time, many in Norway called for the active mandate to be rescinded from its fund manager, NBIM. But the Ministry of Finance did not act in haste. I was one of three researchers asked to analyze the fund's investments as part of building

³⁷ Creswell, J., "Pensions Find Riskier Funds Fail to Pay Off," *New York Times*, April 1, 2012,

³⁸ As determined by American CoreLogic, a real estate data company. See Streitfield, D., "A Town Drowns in Debt as Home Values Plunge," *New York Times*, Nov. 10, 2008.

³⁹ See more details in Ang and Kjaer (2011).